OTH REGENSBURG

# Praktikum Regelungstechnik (RTP)

# Versuch T2: Systemidentifikation

Versuch 1

Gruppe 38
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`Dokument-Version: 02`

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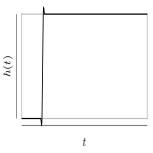
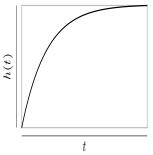
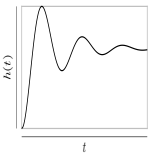
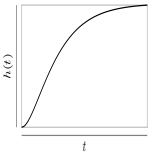
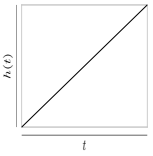
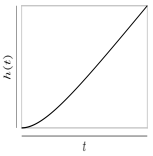
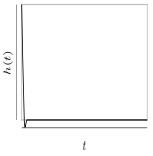
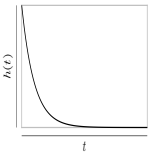
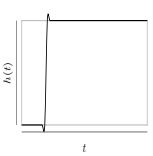
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# 1 Preperations

In preparation for this experiment, the files provided in ELO were downloaded and saved [**RTP˙Experiments**]. In addition, the step responses for various system behaviours were researched:

## 1.1. Different basic system behaviors

Check names of columns again

| **Systemverhalten** | **Übertragungsfunktion** | **Visualization** |
|---|---|---|
| P-Glied ($K_p$) | $G(s) = K_p$ (1.1) |  |
| $PT_1$-Glied ($K_p$, $T_1$) | $G(s) = \frac{K_p}{T_1 s + 1}$ (1.2) |  |
| $PT_2$-Glied ($K_p$, $\omega_0$, $D$) | $G(s) = \frac{K_p}{\frac{1}{\omega_0^2} s^2 + \frac{2D}{\omega_0} s + 1}$ (1.3) |  |
| $PT_2$-Glied für $D \geq 1$ ($K_p$, $T_1$, $T_2$) | $G(s) = \frac{K_p}{(T_1 s + 1)(T_2 s + 1)}$ (1.4) |  |
| I-Glied ($K_I$) | $G(s) = \frac{K_I}{s}$ (1.5) |  |
| $IT_1$-Glied ($K_I$, $T_1$) | $G(s) = \frac{K_I}{s(T_1 s + 1)}$ (1.6) |  |
| D-Glied ($K_D$) | $G(s) = K_D\, s$ (1.7) |  |
| $DT_1$-Glied ($K_D$, $T_1$) | $G(s) = \frac{K_D\, s}{T_1 s + 1}$ (1.8) |  |
| $T_t$-Glied ($K_p$, $T_t$) | $G(s) = K_p\, e^{-T_t s}$ (1.9) |  |

### 1.1.1. P-Glied

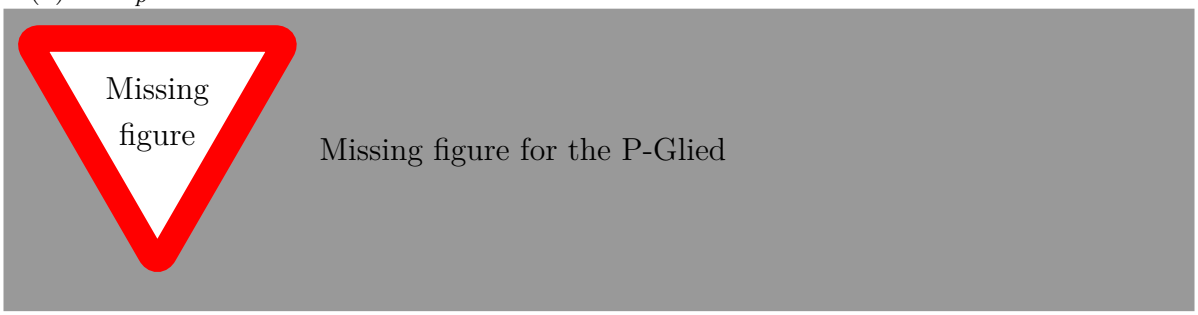
#### Explanation

Check explanation of the P-Glied again, look into script

Proportional-Glied, show a constant behavior in steady state for a constant input. The output is directly proportional to the input, with the proportionality constant being $K_p$. The equation for a P-Glied is given by

$$x_a(t) = K_p x_e(t) \tag{1.10}$$

where $x_a(t)$ is the output and $x_e(t)$ is the input. The transfer function of a P-Glied is $G(s) = K_p$ 1.1.




Missing figure for the P-Glied

#### Parameters needed for characterization

The only parameter needed to characterize a P-Glied is the proportional gain $K_p$. This parameter determines the steady-state gain of the system and how much the output will change in response to a change in the input.

#### Step and Impulse response

The step response of a P-Glied is a step function that reaches a steady state value of $K_p$ times the input step magnitude. The impulse response is an impulse function scaled by $K_p$.

## 1.1.2. PT$_1$-Glied

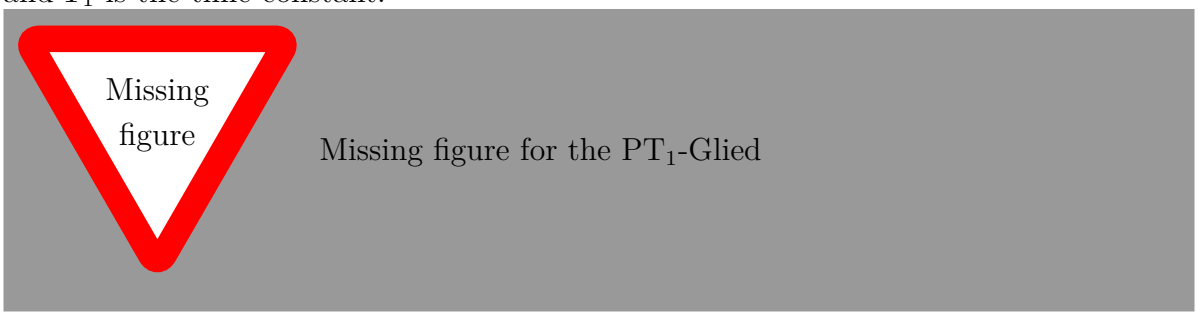
### Explanation

Check explanation of the PT1-Glied again, look into script

The PT$_1$-Glied is a first-order delay system that consists of a proportional element and a first-order part. The transfer function of a PT$_1$-Glied is given by

$$T_1 \dot{x}_a + x_a(t) = K_p x_e(t) \tag{1.11}$$

or equation 1.2, where $K_p$ is the proportional gain and the final value of the function and $T_1$ is the time constant.




Missing figure for the PT$_1$-Glied

### Parameters needed for characterization

For the characterization of a PT$_1$-Glied, the parameters needed are the proportional gain $K_p$ and the time constant $T_1$. The proportional gain $K_p$ determines the steady-state gain of the system, while the time constant $T_1$ determines how quickly the system responds to changes in the input. The System is also characterized by its initial tangent, which is given by $y(t) = \frac{K_p}{T_1} t$. The point at which the initial tangent intersects the steady state value of the step response can be used to determine the time constant $T_1$. $T_1$ is the time it takes for the system to reach approximately 63% of its final value in response to a step input. The threefold multiple of the time constant $T_1$ at which the system reaches 95% of its final value can also be used to characterize the system's response speed.

### Step and Impulse response

The step response of a PT$_1$-Glied is an exponential function that approaches a steady state value of $K_p$ times the input step magnitude, with a time constant of $T_1$. The impulse response is an exponential function scaled by $K_p/T_1$.

## 1.1.3. PT$_2$-Glied

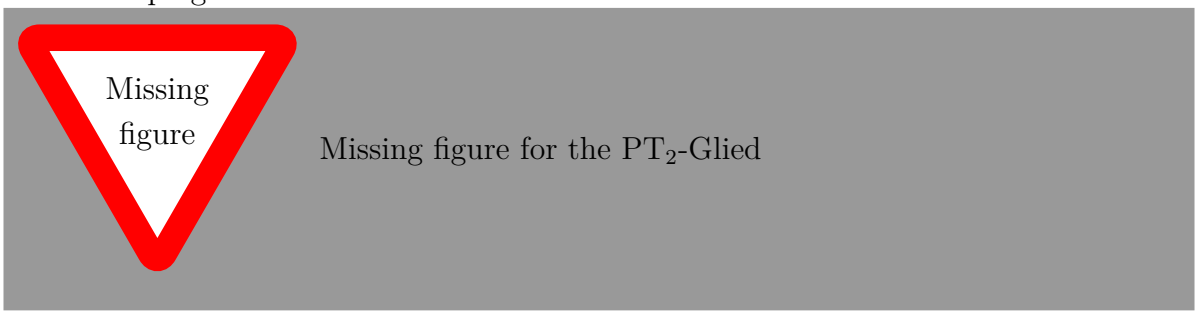
### Explanation

Check explanation of the PT2-Glied again, look into script

A PT$_2$-Glied is a second-order delay system that consists of a proportional element and a second-order part, which appares delayed and oscillatory behavior becuase of its second order differential nature. The transfer function of a PT$_2$-Glied is given by

$$\frac{1}{\omega_0^2}\ddot{x}_a + \frac{2D}{\omega_0}\dot{x}_a + x_a(t) = K_p x_e(t) \tag{1.12}$$

or equation 1.3 , where $K_p$ is the proportional gain, $\omega_0$ is the natural frequency, and $D$ is the damping ratio.




Missing figure for the PT$_2$-Glied

### Parameters needed for characterization

For the characterization of a PT$_2$-Glied, the parameters needed are the proportional gain $K_p$, the natural frequency $\omega_0$, and the damping ratio $D$. The proportional gain $K_p$ determines the steady-state gain of the system, while the natural frequency $\omega_0$ and damping ratio $D$ determine the oscillatory behavior and response speed of the system. The damping ratio $D$ indicates whether the system is underdamped ($D < 1$), critically damped ($D = 1$), overdamped ($D > 1$) or undamped ($D = 0$).
For an underdamped system ($D < 1$), the step response will exhibit oscillations before settling to the steady state value. The natural frequency $\omega_0$ determines the frequency of these oscillations, while the damping ratio $D$ determines how quickly the oscillations decay. For a critically damped system ($D = 1$), the step response will reach the steady state value without oscillating, and it will do so as quickly as possible without overshooting. It is often reached by putting two PT$_1$-Glied in series, which results in the transfer function given by

$$T_1 T_2 \ddot{x}_a + (T_1 + T_2)\dot{x}_a + x_a(t) = K_p x_e(t) \tag{1.13}$$

or

$$h(t) = K_P \left(1 - (1 + \omega_0 t)e^{-\omega_0 t}\right) \tag{1.14}$$

For an overdamped system ($D > 1$), the step response will reach the steady state value without oscillating, but it will take longer to do so than a critically damped system. The Transfer function can be rewritten as

$$h(t) = K_p \left(1 + \frac{1}{\lambda_1 - \lambda_2}(\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t})\right) \tag{1.15}$$

where $\lambda_1$ and $\lambda_2$ are the eigenvalues of the characteristic equation of the system, given by

$$\lambda_{1,2} = -D\omega_0 \pm \omega_0\sqrt{D^2 - 1} \tag{1.16}$$

The time constants of the system can be determined from these roots, and they indicate how quickly the system responds to changes in the input. For an undamped system ($D = 0$), the step response will exhibit sustained oscillations at the natural frequency $\omega_0$.
For the case $0 < D < 1$, the transfer function can be rewritten as

$$h(t) = K_p \left(1 - \frac{e^{-\omega_0 D t}}{\sqrt{1 - D^2}} \sin(\omega_d t + \phi)\right) \tag{1.17}$$

The Periodic time of the oscillations can be determined from the damped natural frequency

$$T_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_0\sqrt{1 - D^2}} \tag{1.18}$$

The Transferfunction can be caracterized by the upper and lower envelope of the oscillations, which are given by

$$h^{\pm}(t) = K_p(1 \pm \frac{e^{-} D\omega_0 t}{\sqrt{1 - D^2}}) \tag{1.19}$$

The over swing time $t_{max}$, which is the time at which the first maximum of the step response occurs, can be determined from the equation

$$t_{max} = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_0\sqrt{1 - D^2}} \tag{1.20}$$

The overshoot $h_{max}$, can be determined from the equation

$$h_{max} = h(t_{max}) = K_p \left(1 + e^{-\pi D/\sqrt{1-D^2}}\right) \tag{1.21}$$

The normalized overshoot value $\ddot{u}$ can be determined from the equation

$$\ddot{u} = \frac{h_{max} - h(\infty)}{h(\infty)} = e^{\frac{-\pi D}{\sqrt{1-D^2}}} \tag{1.22}$$

which can be used to determine the damping ratio $D$ from the overshoot of the step response. Given by the equation

$$D = \frac{|\ln(\ddot{u})|}{\sqrt{\pi^2 + \ln^2(\ddot{u})}} \tag{1.23}$$

**Step and Impulse response**

explain two more system behaviors

# 2 Experiment T2: System identification

## 2.1. System 0

### 2.1.1. Experimentaly Identified system parameters

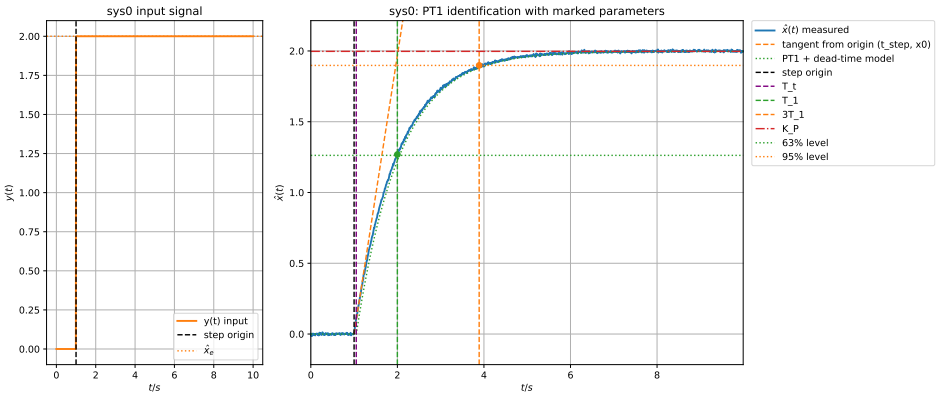



Abbildung 2.1.:

add parameters

### 2.1.2. Documentation of the results

#### Essential Commands

add essential commands for PT1 identification

#### Differential Equation

add differential equation for PT1 system

#### Transfer Function

add transfer function for PT1 system

### 2.1.3. Realworld Example

A practical example of such a PT1 system with impulsive excitation is the temperature response of a small metal body to a brief heat flux. If, for example, a metal rod is briefly excited by a heat source (e.g. hot air or a laser) — comparable to a Dirac impulse — the body heats up abruptly and then cools down again via heat conduction and dissipation to the surroundings. The thermal behaviour follows a PT1 model: the temperature rises briefly due to the pulse (corresponding to the step in the output signal) and then falls back to the original level exponentially with a characteristic time constant. The time constant depends on the material properties (heat capacity, thermal conductivity) and the environmental conditions (e.g. convection, radiation). Such system responses frequently occur in sensor technology, materials analysis or in the case of short control pulses in electronic or thermal systems.description of the respective input and output variables.

### 2.1.4. Comparison of given System with the response of the identified system

add short text about differences of the ideal system to the messured PT1 system

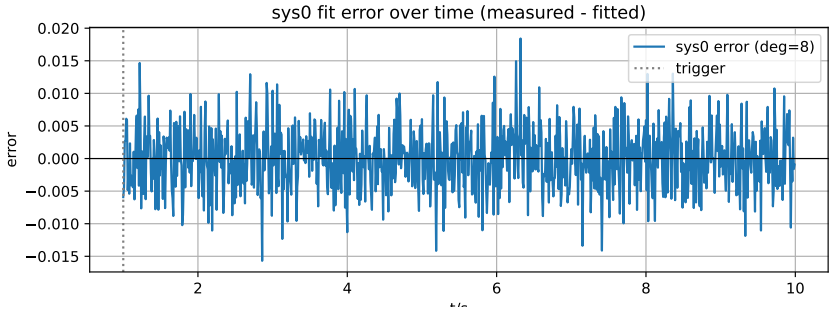



Abbildung 2.2.:

#### Deviations of the System

# 2.2. System 1

## 2.2.1. Experimentaly Identified system parameters

add parameters

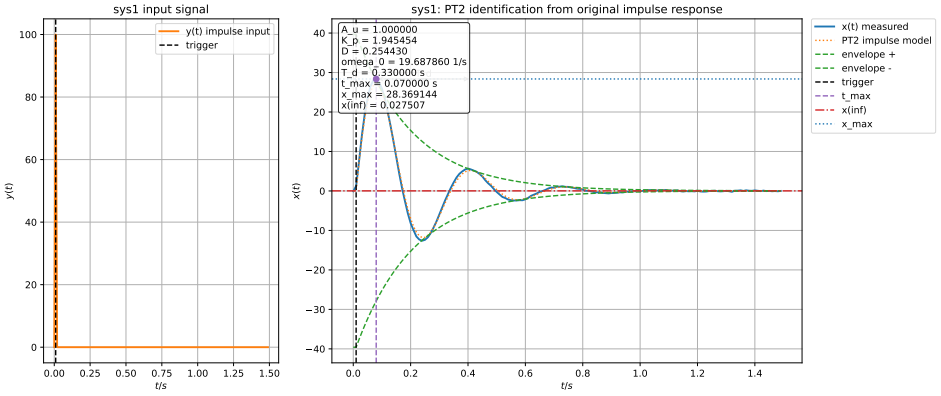



Abbildung 2.3.:

## 2.2.2. Documentation of the results

### Essential Commands

add essential commands for PT2 identification

### Differential Equation

add differential equation for PT2 system

### Transfer Function

add transfer function for PT2 system

## 2.2.3. Realworld Example

add realworld example for PT2 system, look into script

description of the respective input and output variables.

### 2.2.4. Comparison of given System with the response of the identified system

add short text about differences of the ideal system to the messured PT2 system

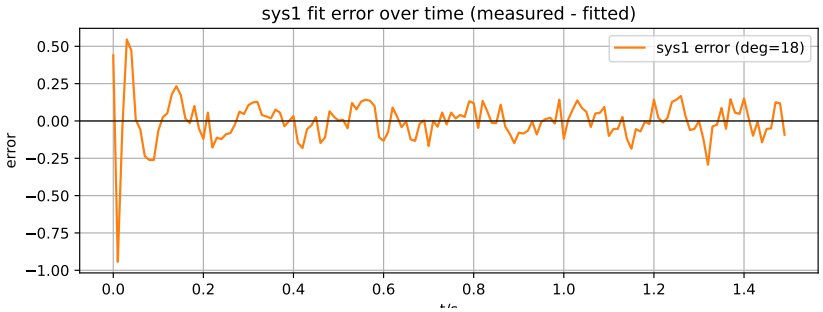



Abbildung 2.4.:

#### Deviations of the System

# 2.3. System 2

## 2.3.1. Experimentaly Identified system parameters

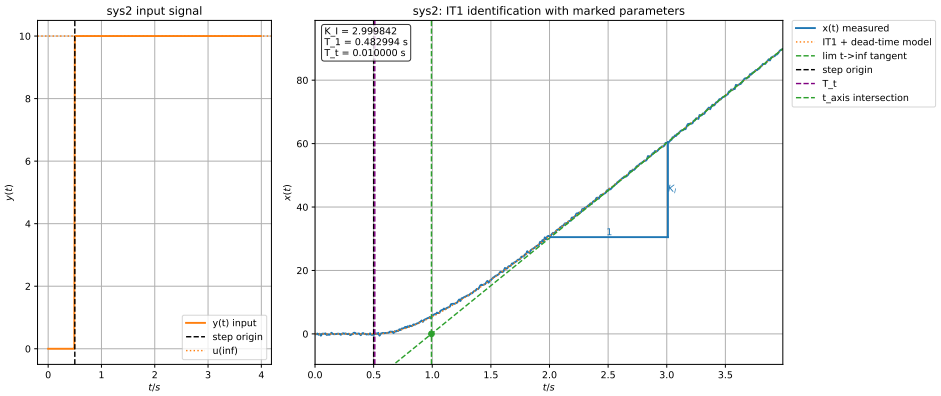



Abbildung 2.5.:

add parameters

## 2.3.2. Documentation of the results

### Essential Commands

add essential commands for IT1 identification

### Differential Equation

add differential equation for IT1 system

### Transfer Function

add transfer function for IT1 system

## 2.3.3. Realworld Example

A clear example of an IT1 element with an upstream dead time is a water tank with a constant inflow and proportional outflow. The inflow corresponds to the input signal, and the water level to the output signal. The storage results in integrative behaviour, whilst the proportional outflow causes first-order damping — typical of an IT1 system. In the experimental setup under consideration, a dead-time element was connected upstream of the IT1 element, causing a time delay in the input signal. This means that the

system output only reacts to a change after the dead time has elapsed. Such combinations of dead time and IT1 behaviour frequently occur in real-world processes, for example in sluggish piping systems or delayed control variable transmissions. description of the respective input and output variables.

### 2.3.4. Comparison of given System with the response of the identified system

add short text about differences of the ideal system to the messured IT1 system

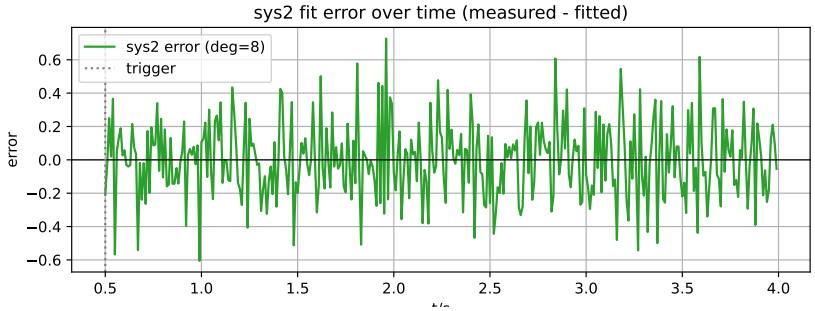



Abbildung 2.6.:

#### Deviations of the System

# A Appendix Frontmatter

## A.1. Declerations

add declerations

## A.2. List of Figures

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## A.4. List of Variables and Constants used in this report

# B Appendix Code and Resources

## B.1. Code used for the analysis of the Preperations and the Experiment

add code for the analysis of the preperations, look into jupyter notebook

# C Bibliography